



CROSS-INSTANCE GAUSSIAN SPLATTING REGISTRATION VIA GEOMETRY-AWARE FEATURE-GUIDED ALIGNMENT

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Motivation & Problem

Goal: category-level 3DGS alignment. Given two independently reconstructed Gaussian Splatting models, estimate a global similarity transform

$$T = (s, R, \mathbf{t}) \in \text{Sim}(3)$$

that aligns the source model $\mathcal{G}_1$ to the target model $\mathcal{G}_2$.

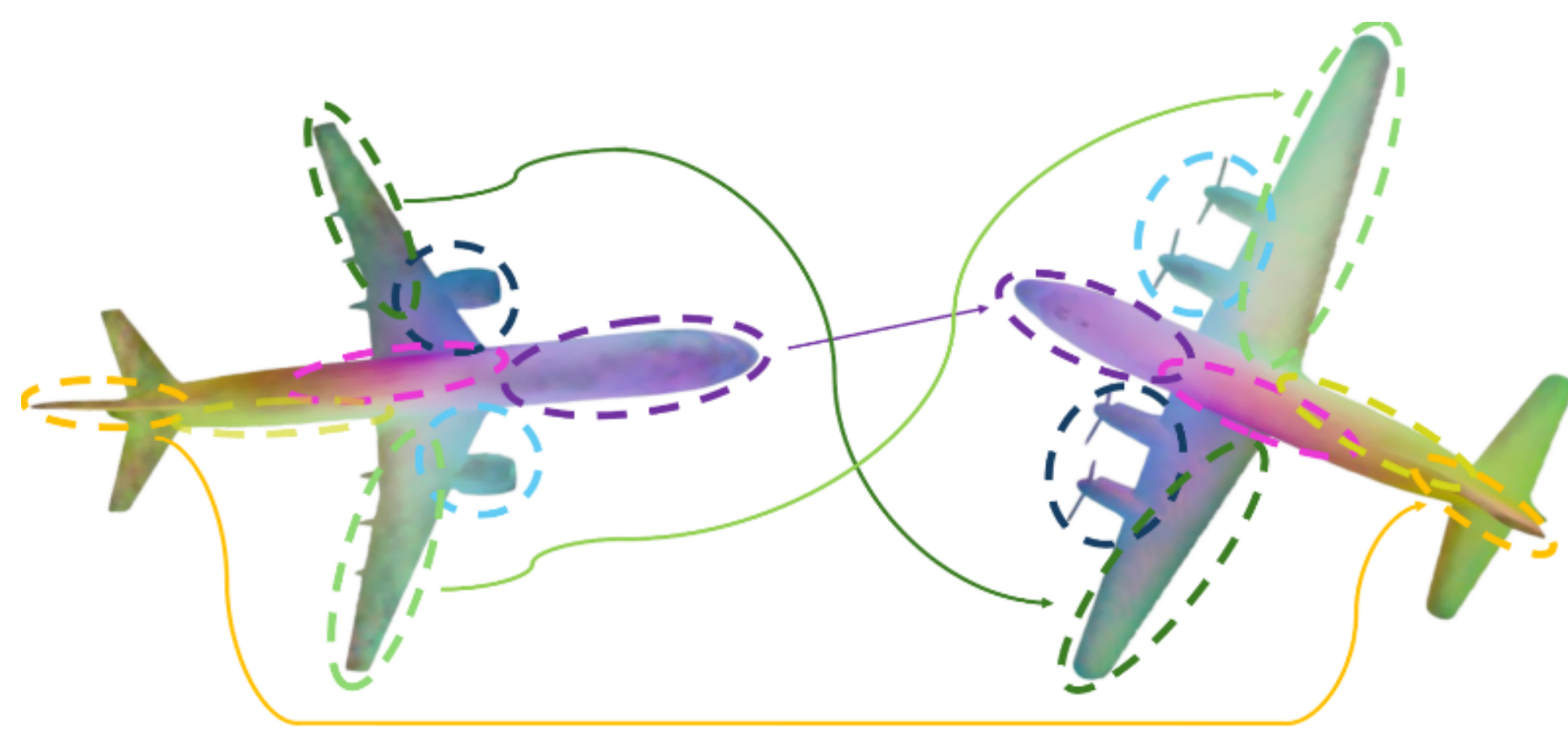
Why it matters

- Align different object instances.
- Enable synchronized rendering.
- Support geometrically consistent object replacement.

Why it is hard

- Arbitrary frames and scales.
- Shape variation across instances.
- Ambiguous or missing correspondences.

Geometry-Aware Semantic Features



Geometry-aware semantic features capture both semantic similarity (e.g., wings, engines) and spatial distinctions (e.g., left/right, front/back).



Input

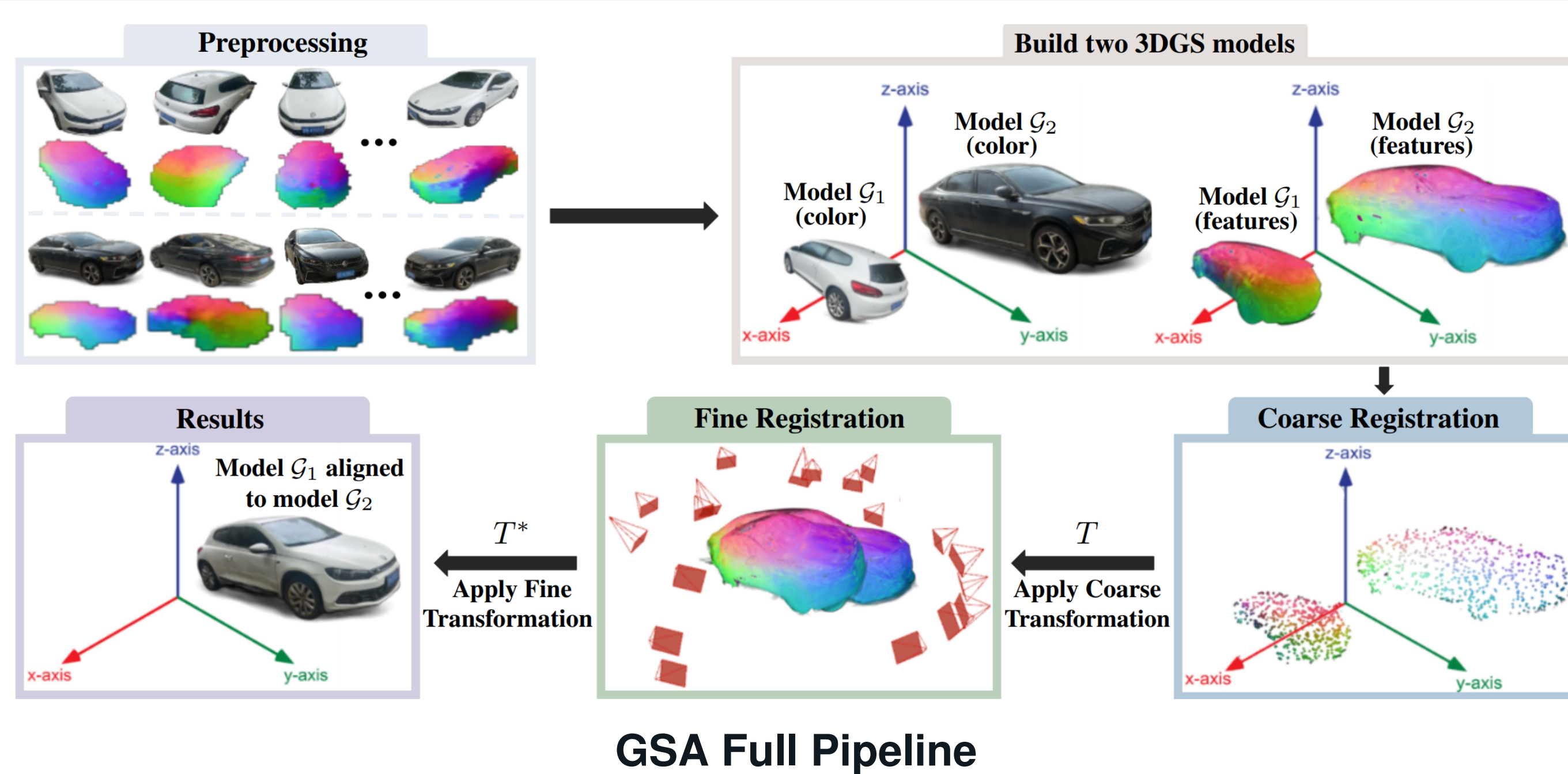
RGB Render

Input Feat.

Rendered Feat.

Low-resolution 2D semantic-geometric features are lifted into 3DGS and rendered as high-resolution feature maps.

GSA Overview



GSA Full Pipeline

Coarse: Feature-Guided Sim(3) Solver

Idea: extend ICP-like absolute orientation with semantic-geometric pruning, then solve a closed-form Sim(3) update.

1. Feature-guided candidate set

$$Q_i = \{ \mathbf{q}_j \in \mathcal{G}_2 \mid \|\mathbf{f}_i - \mathbf{f}_j\|_2 \leq \tau_f \}$$

2. Spatial nearest neighbor after pruning

$$\mathbf{q}_i^{(k)} = \arg \min_{\mathbf{q} \in Q_i} \|\mathbf{p}_i^{(k)} - \mathbf{q}\|_2$$

3. Closed-form similarity update

$$T^{(k)} = \arg \min_{T \in \text{Sim}(3)} \sum_i \left\| T(\mathbf{p}_i^{(k)}) - \mathbf{q}_i^{(k)} \right\|_2^2$$

The coarse stage is robust to poor initializations, including large rotations, scale gaps, and cross-instance shape variation.

Fine: From Inverse Radiance Fields to MVFC

Classical inverse radiance-field pose estimation [2]

$$\min_{C \in \text{SE}(3)} \left\| \text{Rend}_{\text{rgb}}(\mathcal{G}, C) - I_t \right\|_2^2$$

Extension to multi-view field-to-field alignment

$$\min_{T \in \text{Sim}(3)} \sum_{k=1}^N \left\| \text{Rend}_{\text{rgb}}(T\mathcal{G}_1, C_k^*) - \text{Rend}_{\text{rgb}}(\mathcal{G}_2, C_k^*) \right\|_2^2$$

Adaptation to cross-instance feature-field alignment

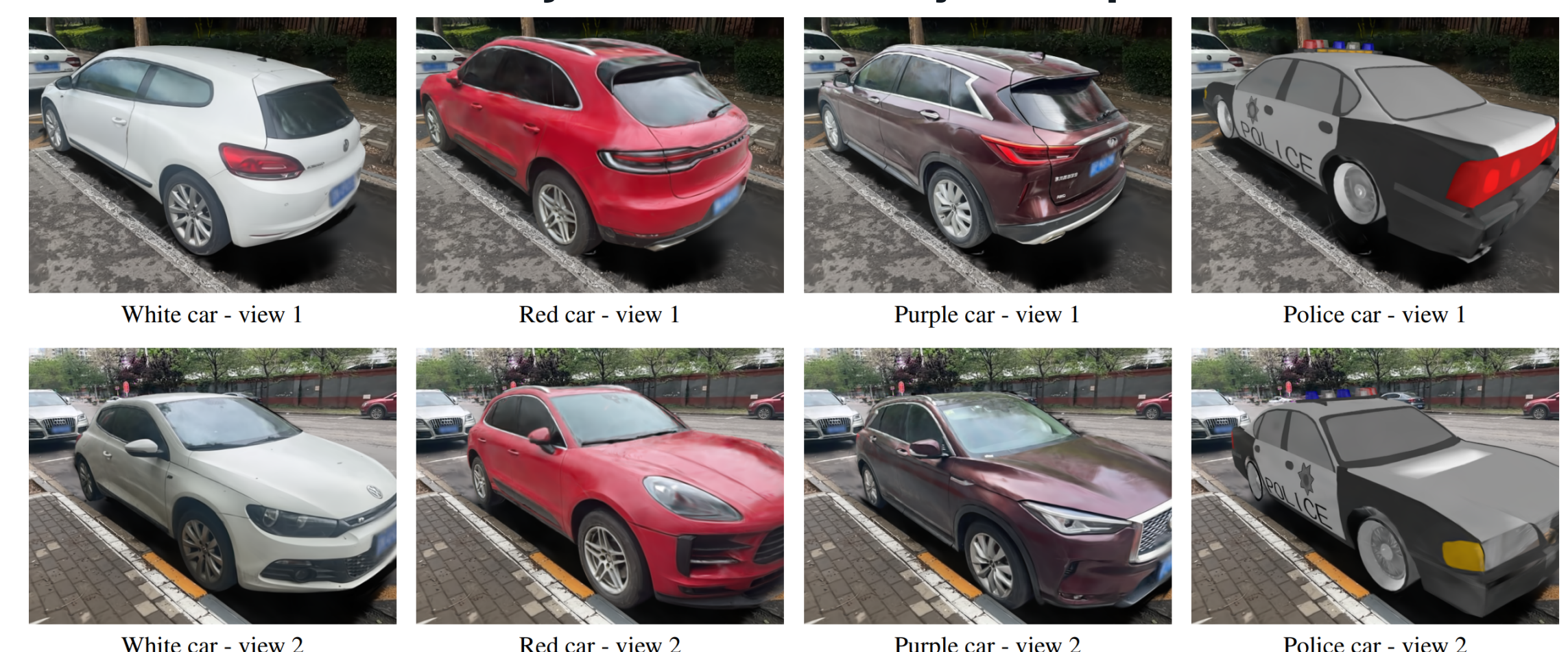
$$\mathcal{L}_{\text{MV-FC}} = \sum_{k=1}^N \left\| \text{Rend}_f(T\mathcal{G}_1, C_k^*) - \text{Rend}_f(\mathcal{G}_2, C_k^*) \right\|_2^2$$

$$T^* = \arg \min_{T \in \text{Sim}(3)} \mathcal{L}_{\text{MV-FC}}$$

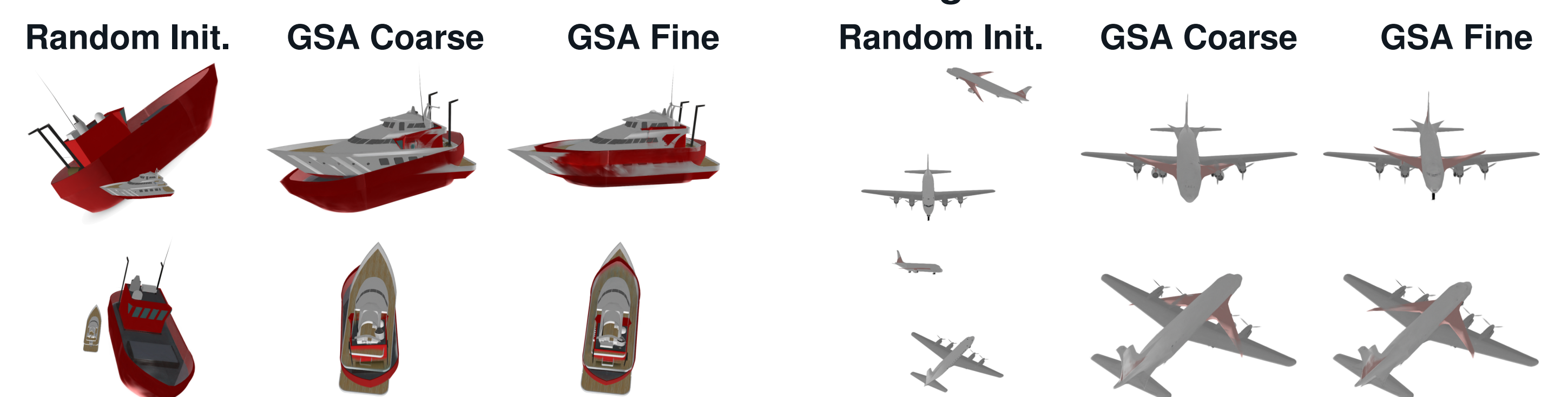
The fine stage refines the coarse alignment by rendering both feature-augmented 3DGS models from multiple target views and optimizing the global Sim(3) transformation.

Qualitative Results

Geometrically Consistent Object Replacement



Cross-Instance 3DGS Registration



Quantitative Results

Same-Object 3DGS Registration

Method	RRE (°) ↓	ATE ↓
FGR	51.32	0.33
REGTR	125.34	0.38
DReg-NeRF	10.37	0.04
GaussReg	18.60	0.07
GSA Coarse	3.13	0.01
GSA Coarse+Fine	0.36	0.001

Cross-Instance Alignment on Synthetic Objects

Method	Airplane	Bus	Boat	Car	Chair	Motor.
FGR	90.78	94.98	86.60	89.32	91.53	82.06
REGTR	90.00	77.69	90.98	96.98	94.75	86.97
GaussReg	136.32	138.51	126.64	121.36	155.03	120.14
GSA Coarse	2.97	4.74	23.97	3.70	14.17	3.13
GSA Coarse+Fine	2.45	2.41	22.68	1.71	10.86	4.05

Cross-Instance Alignment on Real Objects

Method	CO3Dv2 toyplane	CO3Dv2 bicycle	CO3Dv2 chair	3D Real car
GaussReg	121.77	122.27	125.66	107.66
GSA Coarse	6.44	4.60	33.12	1.93
GSA Coarse+Fine	5.22	9.27	22.67	1.34

Takeaway: GSA recovers full Sim(3) alignment across different object instances and strongly outperforms geometry-only or same-instance baselines.

QR Code & References



Scan for paper, code, video, and additional results.

[1] Mariotti et al. *Improving Semantic Correspondence with Viewpoint-Guided Spherical Maps*. CVPR 2024.

[2] Yen-Chen et al. *INeRF: Inverting Neural Radiance Fields for Pose Estimation*. IROS 2021.